MACHINE LEARNING · NAUKMA

Brainrot Detector

Can a neural network tell **Italian brainrot** memes from everything else?

- ✓ **Yes**, it can tell them apart from photos
- ! **Not yet** from other AI-generated art



Team

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THE DATA

Every brainrot image is AI-made, so we added ordinary AI art to keep the model honest



Brainrot 4 982



Ordinary AI art 500



Photos 4 500

PATTERNS IN THE DATA

Brainrot is more colourful. Not brighter.



Brainrot



Photo

Colourfulness

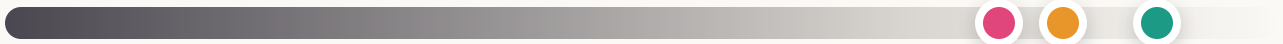
CLEAR DIFFERENCE



Photos	AI art	Brainrot
79	92	116

Brightness

ALMOST IDENTICAL

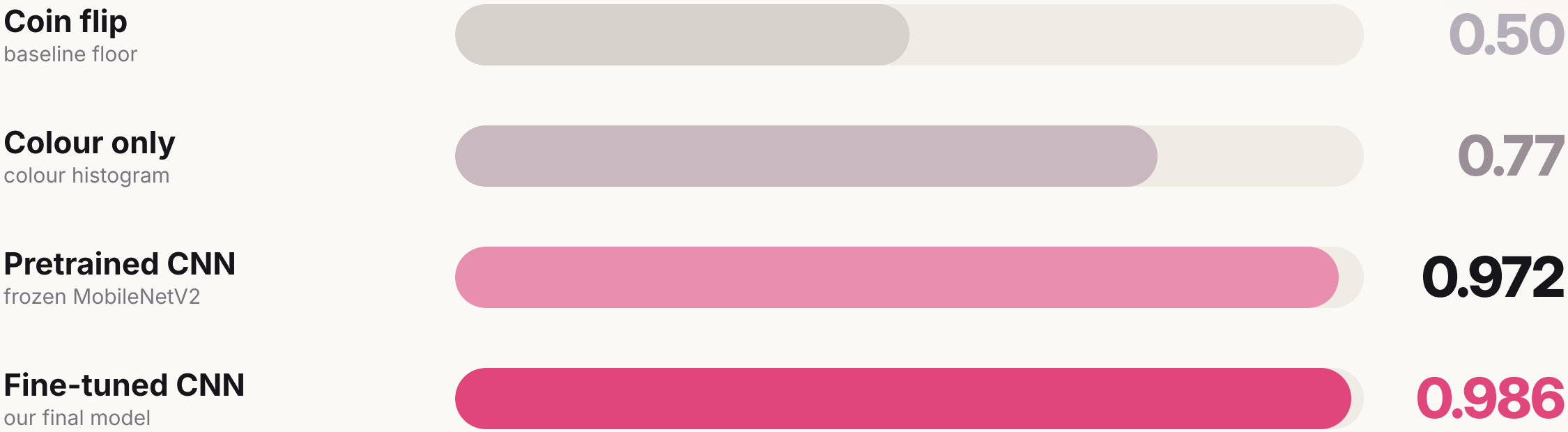


all three
123–133

Average per image, scale 0–255 (zoomed to 60–140) · Size, crop and file quality were made identical for all images, so they can't give the answer away.

WHAT WE BUILT

Four models, each better than the last



Score = how reliably brainrot is ranked above everything else (ROC-AUC; 1.0 = perfect, 0.5 = coin flip).
Measured once on 1 498 images the models never saw.

RESULTS

Great with photos. Fooled by AI art.




1.5%

of photos wrongly called brainrot

✓ WORKS




4.7%

of brainrot images missed

✓ WORKS




38.7%

of ordinary AI images wrongly called brainrot

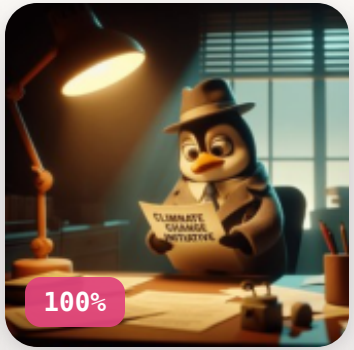
× WEAK SPOT

WHY

It learnt a shortcut: “cute AI creature = brainrot”

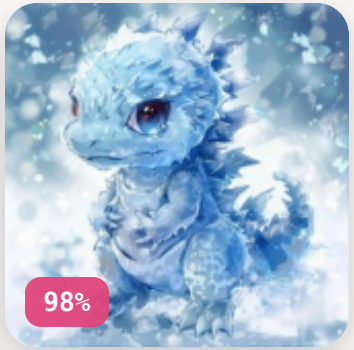










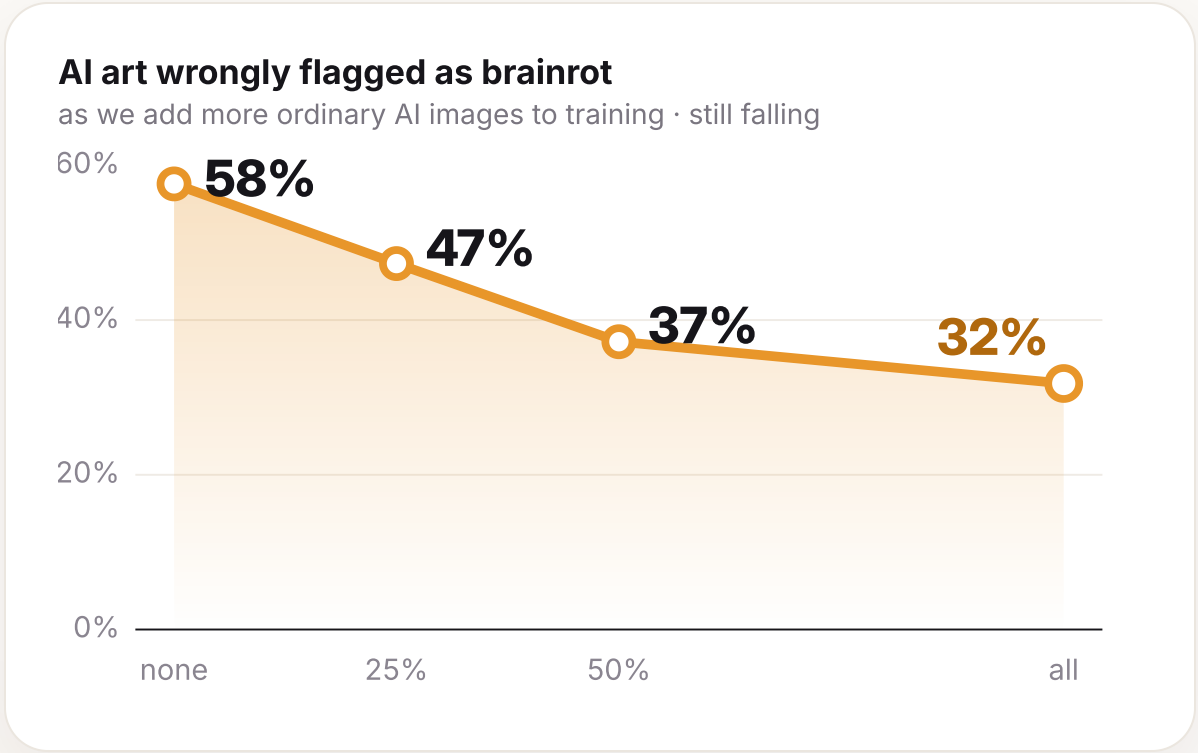


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ordinary AI images marked as brainrot, with near-100% confidence. None of these is a brainrot meme.

WHAT IT MEANS

The fix: show it more ordinary AI art



58% → 32%

Adding ~340 ordinary AI images to training almost halved these false alarms, and the curve has not flattened yet.



More of these = fewer mistakes

CONCLUSION

A reliable brainrot-vs-photo filter, one data fix away from more

0.986

score against photos; only 1.5% false alarms. **Works today.**

39%

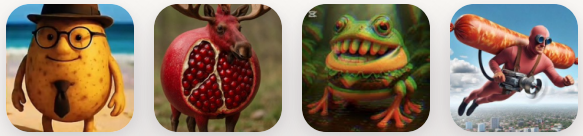
of ordinary AI art still fools it: it learnt **“cute AI creature”**.

Next

collect more ordinary AI images. Error already fell **58% → 32%**.

Thank you!

Questions?



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